

Practice Exam 2

Do not open this exam until instructed to do so.

- You have 75 minutes to complete this exam
- You may use a calculator; you may **not** use any other device (cell phone, etc.)
- You may consult one page of notes (both sides); you may not use books, notebooks, etc.
- Show your work

I understand that the honor code applies: I will not lie, cheat, or steal to gain an academic advantage, nor tolerate those who do.

Signature

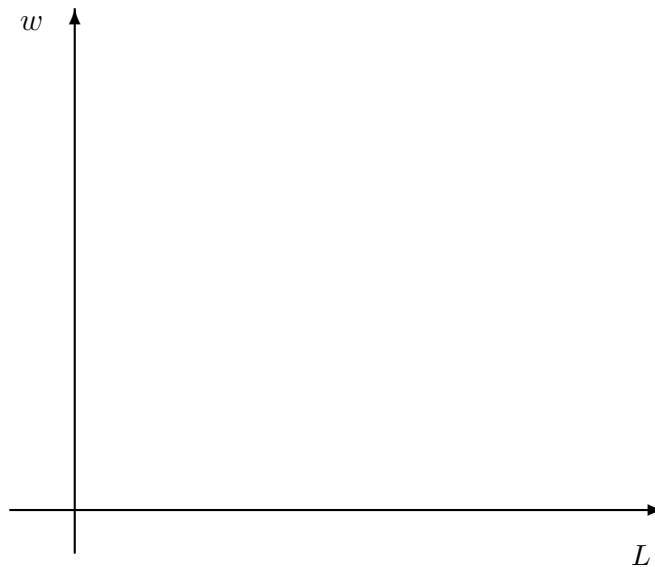
Printed Name

For each question below, **write the letter of the most correct answer to the left of the question.**

1. (3 pts.) The labor force participation rate is:
 - A. The ratio of employees to the working-age population
 - B. The ratio of employees and the unemployed to the working-age population
 - C. The ratio of employees and the unemployed to the population
 - D. Usually lower than the employment rate
2. (3 pts.) Hyperinflations cannot occur in the absence of:
 - A. Wars
 - B. Social crises
 - C. Adverse supply shocks
 - D. Rapid money growth
3. (3 pts.) The Beveridge curve is an indicator of:
 - A. The efficiency of the labor market
 - B. The willingness of workers to supply labor
 - C. The demand for labor
 - D. All of the above
4. (3 pts.) Indicators that are more volatile than GDP include all the following, *except*:
 - A. Inventories
 - B. Sales of consumer durable goods
 - C. Employment
 - D. Shipments of business equipment
5. (3 pts.) An economy's employment separation rate is the number of people who have left a job in the period divided by
 - A. The number of people employed
 - B. The working-age population
 - C. The labor force
 - D. The number of people unemployed
6. (3 pts.) The cross-correlation function/figure determines:
 - A. If a variable leads or lags the business cycle
 - B. If a variable is pro-, counter-, or acyclical
 - C. If a variable is more or less volatile than GDP
 - D. Both A. and B.

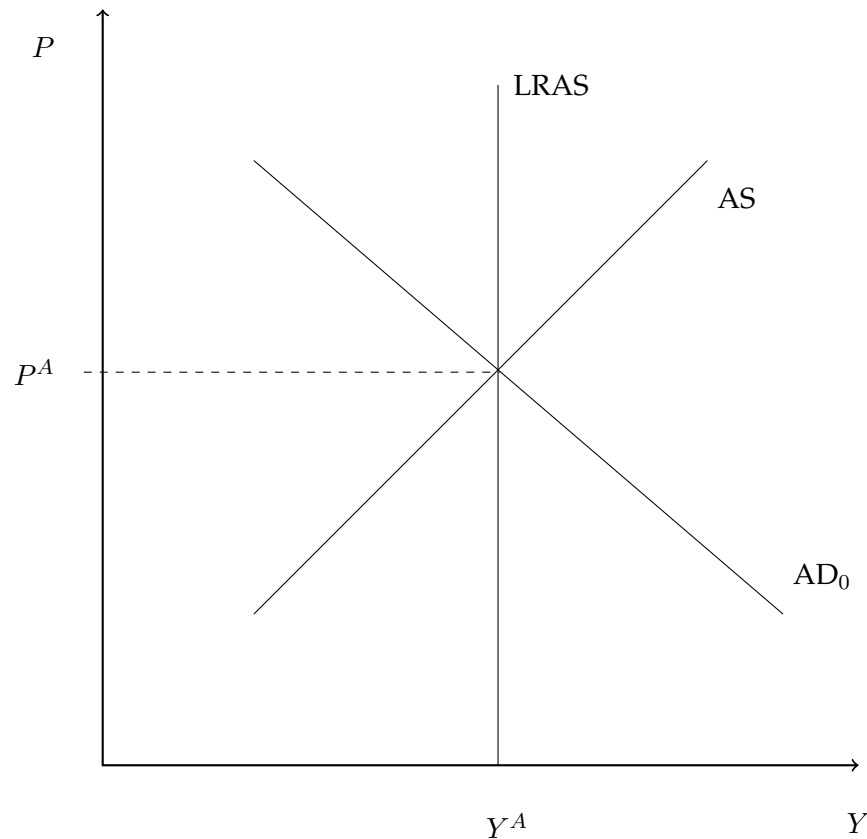
7. (3 pts.) Wage stickiness accounts for:
- A. The vertical slope of the long-run aggregate supply curve
 - B. The downward slope of the aggregate demand curve
 - C. The upward slope of the short-run aggregate supply curve
 - D. The upward slope of the Beveridge curve
8. (3 pts.) When the Fed keeps the federal funds rate persistently above the Taylor Rule rate, we expect:
- A. A recession and disinflation
 - B. A boom and rising inflation
 - C. A declining risk of deflation
 - D. None of the above
9. (3 pts.) In the labor market dynamics model, the unemployment rate evolves according to $u_{t+1} = u_t + s(1 - u_t) - au_t$, where s is the separation rate and a is the accession rate. If s is held fixed and a increases, all of the following are true *except*:
- A. The average length of an unemployment spell decreases.
 - B. The steady state unemployment rate decreases.
 - C. The average length of time at a job increases.
 - D. The labor market returns to a steady state.
10. (3 pts.) When expected inflation rises, the real interest rate on previously issued debt:
- A. Falls
 - B. Is unaffected
 - C. Rises
 - D. Rises only if the nominal interest rate is indexed to inflation

12. **Payroll taxes.** You have been asked to analyze the distortionary effects of wage taxes. The demand function for labor is $L^D(w) = (\frac{3}{2}w)^{-3}$ and the labor supply function is $L^S(w) = w^2$.
- a. (6 pts.) On the axes below, sketch a supply and demand diagram for the labor market. Compute the equilibrium wage and the quantity of labor supplied. Label the equilibrium wage and the quantity of labor supplied in the figure below.



- b. (9 pts.) Suppose we impose a 5 percent payroll tax, which is paid by employers. Compute the pre-tax wage, the after-tax wage, and the amount of labor hired in equilibrium. [Hint: If w is the wage earned by an employee (the after-tax wage) then the pre-tax wage paid by the employer is $w \times 1.05$.]
- c. (2 pts.) Every Republican candidate in the coming election advocates for reducing the tax on labor income. Given your analysis, why would a politician be in favor of decreasing payroll taxes?

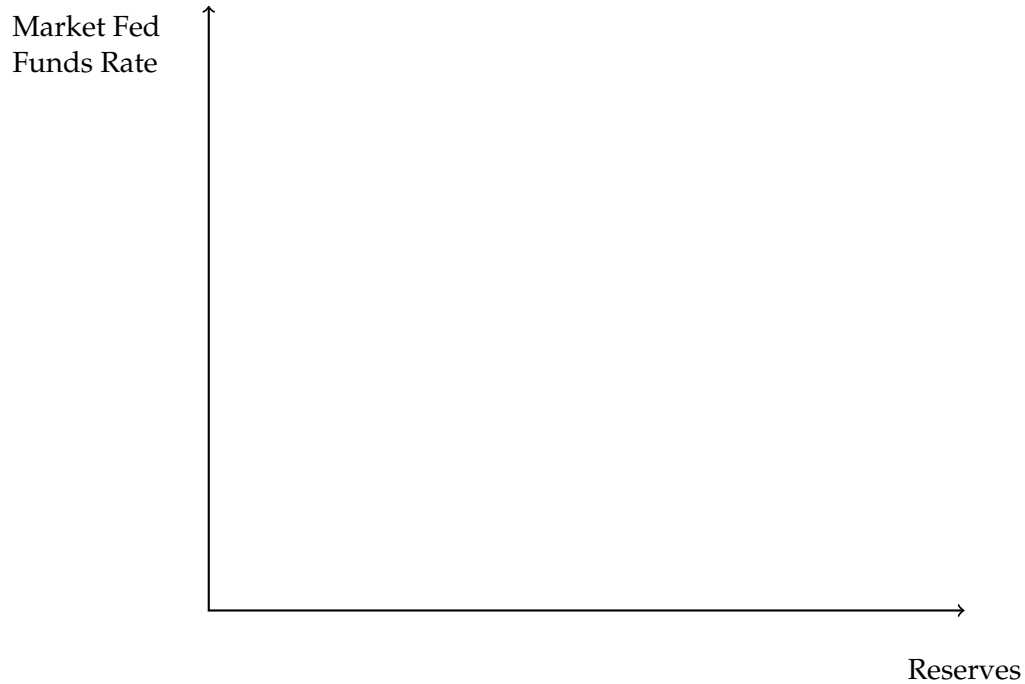
13. **Stabilization policy.** Consider an economy at the equilibrium (Y^A, P^A) below.



- a. (3 pts.) Intensifying deflation in Europe and Japan has sharply diminished confidence among U.S. firms and households. On the graph above, show the effects on the U.S. economy of the fall in confidence. Label the new short-run equilibrium (Y^B, P^B) .

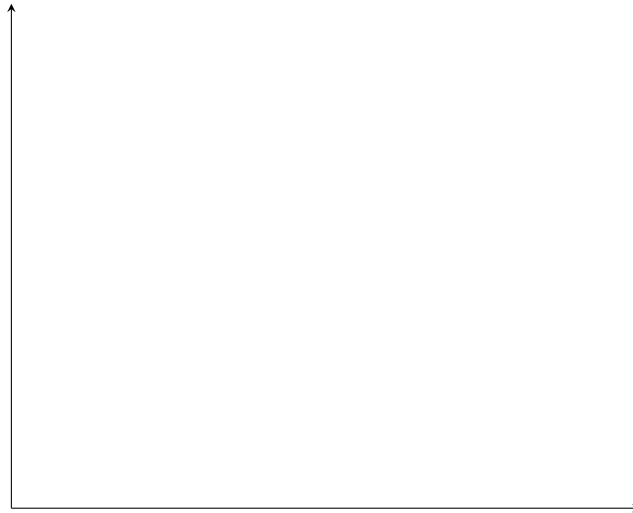
- b. (3 pts.) Investors wonder how the Fed will react. The U.S. inflation rate is 0.6 percent, and the Congressional Budget Office estimates that current real GDP is 5.0 percent below potential GDP. Use the Taylor rule to predict the federal funds target rate. Assume the Fed's inflation target is 2 percent, the long-run real interest rate is 2 percent and the sensitivity parameters (or coefficients) for deviations of inflation from its target and for deviations of output from its potential are both 0.5.
- c. (3 pts.) Why can't the Fed implement this policy target? Be specific.

- d. (4 pts.) With the market federal funds rate at zero, the Fed engages in quantitative easing (QE) to stimulate the economy. On the graph below, draw and label the supply and demand for reserves prior to QE, and label the equilibrium level of reserves as Q^A . Show the impact of the Fed's QE in your figure. Label the post-QE equilibrium level of reserves as Q^B .



- e. (2 pts.) In words, what operations does the Fed conduct to move from Q^A to Q^B in part d? Be specific.

- d. (2 pts.) The Beveridge Curve relates the unemployment rate to the vacancy rate. On the axis below, draw a stylized Beveridge Curve for the U.S. economy before the 2007 recession. Be sure to label both axes.



- e. (6 pts.) Suppose that the economy moves along the Beveridge curve to a point with a lower unemployment rate. What is happening to the accession rate, and why?
- f. (7 pts.) Unemployment in the construction sector spiked in the last recession, and demand for construction employees remains low. At the same time, it is difficult for health care firms to find workers with the appropriate certification. Economists are concerned that the job matching efficiency of the U.S. economy has declined and will remain low for years to come.

In the figure above, show how such a long-term decline of matching efficiency alters the Beveridge Curve. In the space below, explain why.

Extra Space

Clearly label the question number, and leave a reference to this page near the question.